

REVIEW

When Ozempic's signal fades

REFERENCES

31 verified

STYLE

Popsci

TOKENS

≈6.9k

VERIFICATION

Crossref

COMPILED

28 Aug 2026

Stopping Ozempic is usually not a dramatic withdrawal event. It is a gradual return: appetite often grows louder, weight commonly comes back, and the improvements written into blood tests can fade with it—but an average curve is not a personal forecast.

Rubino's trial investigators created the cleanest version of a question that usually unfolds messily in real life. After everyone had spent months taking semaglutide, they kept some participants on the weekly injection and switched the others to placebo. Both groups still received lifestyle support. The people who stopped began gaining weight while those who continued lost more.¹

That split is the central fact behind the worry about stopping Ozempic, a brand of semaglutide. Yet it is not the whole answer. The strongest withdrawal trials mostly studied the higher dose used for obesity in people without diabetes, while many people taking Ozempic use a lower dose to manage diabetes. To understand what your body may do after the last injection, the science has to follow a sequence: the drug signal fades, hunger and eating can change, weight follows, and metabolic markers often change in the same direction. The question is how much of that sequence reaches any one person.^{2,3,4}

01 THE LAST INJECTION KEEPS SPEAKING

Overgaard and colleagues traced semaglutide through the body and found that its plasma concentration falls by half only after roughly a week. That means the final shot does not create a pharmacological cliff; the signal dwindles across several weeks.^{5,6}

What is fading is the action of a GLP-1 receptor agonist, a drug that mimics a gut hormone signal involved in appetite and glucose control. The body is not being shocked by a missing dose so much as released from an ongoing effect. Common nausea, vomiting, diarrhoea and constipation generally track active treatment and dose escalation, so they tend to ease rather than transform into a separate withdrawal syndrome. Wharton's pooled safety analysis found fewer new stomach-and-bowel events among people switched away from semaglutide, although no trial followed every symptom from the precise moment of the final injection.^{7,8,9}

The sequence is easiest to see as a fading signal with several delayed consequences, as shown in Figure 1.

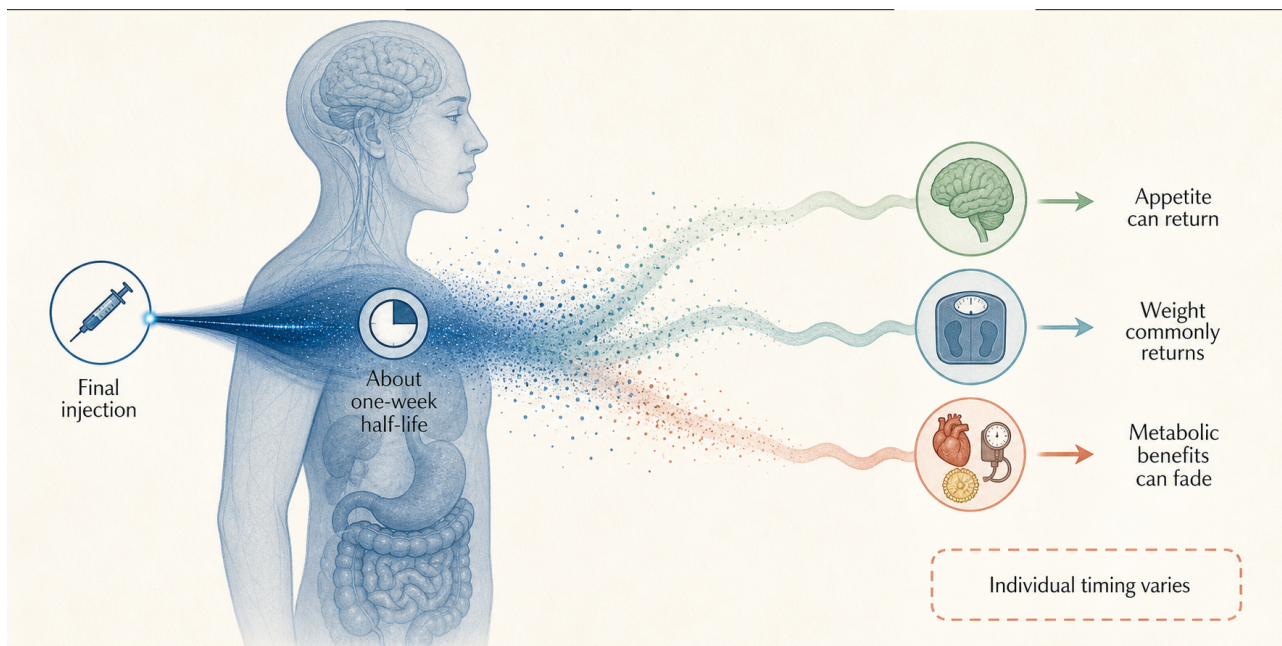


Figure 1. The signal fades before its downstream effects fully show. Semaglutide exposure declines over weeks; reduced appetite and food intake can loosen next, followed by weight regain and erosion of some blood-pressure, glucose and lipid improvements. The order is a synthesis of human pharmacology, feeding studies and withdrawal trials—not a schedule for one person's body.^{5,6,1,2,3}

02 HUNGER GETS ITS OLD VOLUME BACK

Blundell's team brought people with obesity into controlled feeding sessions and watched semaglutide turn down both hunger and how much food they chose to eat. Friedrichsen's team later found the same pattern at the obesity-treatment dose: participants felt fuller and ate about a third less at a test meal.^{6,10} Oral semaglutide studies led by Gibbons and Gabe supplied another version of the same result.^{11,12}

Those experiments did not randomize people to stop and then measure every sandwich. Still, they point to the most direct explanation for regain: when the drug concentration falls, its appetite-dampening effect can fall too. Your stomach may empty somewhat more slowly early in treatment, but longer studies did not find a durable slowdown large enough to explain the sustained weight change. Hunger and eating, not a permanently sluggish stomach, appear to carry most of the story.^{10,12,9}

Damhof and colleagues therefore describe the early post-stop period as a biological transition rather than evidence of weak willpower. But the measurements remain lop-

sided: scientists have carefully measured appetite while the drug is active and have far fewer direct readings after it is gone.¹³

03 THEN THE WEIGHT CURVES PART

Rubino's randomized withdrawal trial showed the curves separating within months. The people switched to placebo regained weight even though lifestyle support continued; the people who stayed on semaglutide kept losing. Crucially, the withdrawal group had not returned all the way to its starting weight by the trial's end.¹

Wilding's team watched a different group for a full year after both semaglutide and structured lifestyle counseling ended. On average, participants regained about two-thirds of their earlier loss, but nearly half still kept a clinically meaningful reduction from baseline.² Berg and colleagues later pooled the randomized cessation evidence and found that the amount regained tended to rise with the amount first lost.¹⁴

The two direct semaglutide studies anchor [Figure 2](#); the table keeps the excess precision out of the story.

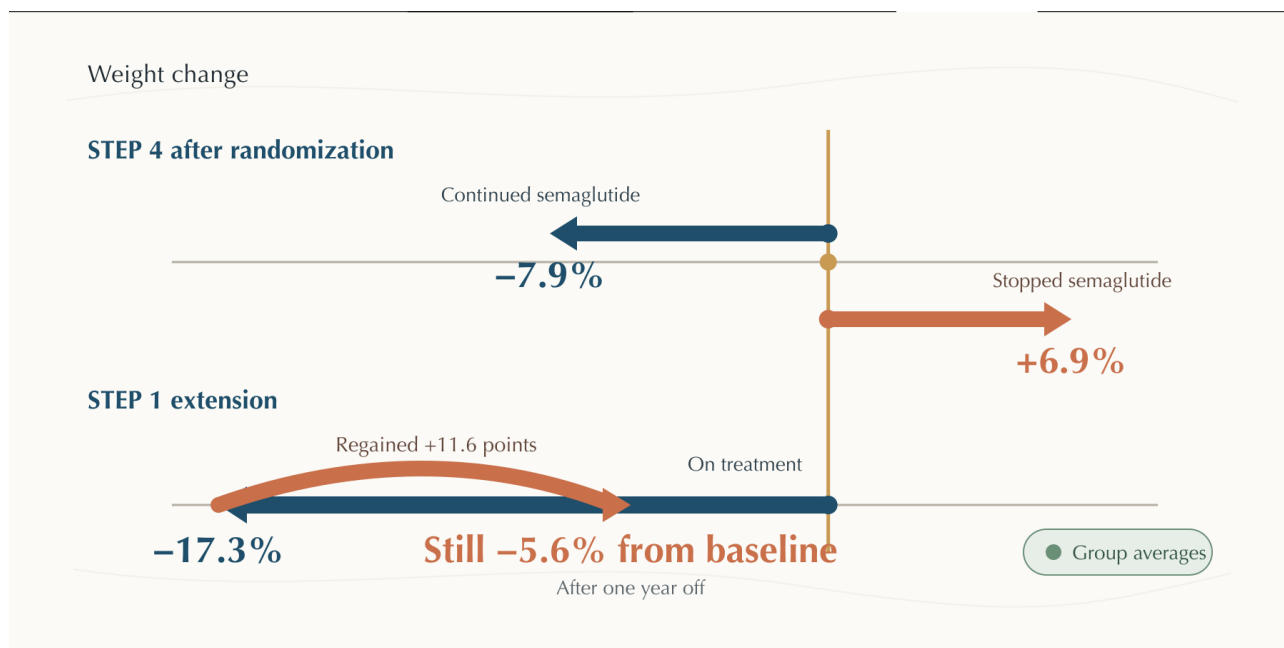


Figure 2. Stopping changed the direction of the average weight curve. In STEP 4, continued treatment and placebo withdrawal moved apart after randomization; in the STEP 1 extension, substantial regain left a smaller residual loss one year after stopping. These are group means from selected trial populations, not promised trajectories.^{1,2,14,15}

EVIDENCE WINDOW	COMPARISON	EXACT WEIGHT RESULT	BOUNDARY
STEP 4 ¹	803 adults; continue 2.4 mg vs stop	-7.9% vs +6.9%; difference -14.8 points (plausible 95% range -16.0 to -13.5)	Lower doses; diabetes
STEP 1 extension ²	327 completers; one year off	-17.3% treated, +11.6 points regained, -5.6% net	Continued counseling
GLP-1 cessation meta-analysis ³	18 trials; obesity and diabetes	Obesity +5.63 kg; diabetes +2.03 kg	Personal trajectory
Weight-management review ¹⁵	37 studies; mixed drugs	Newer incretins +0.8 kg/month	Later values modeled

West's group gathered dozens of cessation studies and estimated faster regain after newer incretin drugs than after older weight-management medicines. Patel and Kolli reached a similar broad conclusion in separate reviews.^{15,16,17} But their longest-range answer is a forecast: the best synthesis had directly observed the newer drugs for no more than about a year after stopping. The body beyond that point is being modeled, not watched.¹⁵

04 BLOOD TESTS FOLLOW THE SCALE

Tzang and colleagues pooled randomized trials that continued measuring people after therapy with these drugs ended. As weight returned, waist size and blood pressure rose, while fasting glucose and HbA_{1c}—a measure of average blood sugar over the preceding months—worsened on average.³ Wilding's follow-up found the same direction: several cholesterol, inflammation and glucose improvements drifted back toward their pretreatment values,

though some advantage remained while weight stayed below baseline.²

For people with diabetes, the balance may look different. The pooled trials found less weight regain than in obesity cohorts but a larger deterioration in that longer-term blood-sugar measure, and the estimate for fasting glucose was too imprecise to rule out little change. Other diabetes-care studies confirm what active semaglutide can control, but they do not turn the pooled cessation result into an Ozempic-specific prediction.^{3,18}

McGowan's prediabetes trial and Wilding's extension add a quieter detail. Many participants moved from prediabetes-range readings to normal glucose regulation during treatment; after withdrawal, that improvement became less common, but it did not vanish for everyone.^{19,2} The more clinically important the glucose-lowering effect was for you, the more consequential its loss may be—even when the scale changes modestly.

05 THE SCALE HIDES THE TISSUE STORY

McCrimmon's team used DXA, an imaging method that estimates body fat and fat-free tissue, while people were taking semaglutide for diabetes. Both fat and lean body mass, the non-fat portion that includes muscle, organs and water, decreased.²⁰ Reviews led by Karakasis, Batsis and Eisa agree that incretin-driven weight loss includes some lean tissue, although their estimates vary with the drug, population and measuring method.^{21,22,23}

What happens to those compartments during regain is much less certain. Ryan, Locatelli and Bhandarkar have all examined the muscle-preservation concern, but repeated

scans after semaglutide withdrawal are scarce. Claims that the returning weight is mainly fat run ahead of direct human measurements.^{24,25,26} The scale can tell you that mass returned. It cannot reveal what tissue rebuilt it.

06 THE AVERAGE IS NOT YOUR FORECAST

So does the trial average tell you what happens next? Gasoyan's team examined health records from people who used injectable semaglutide or tirzepatide in ordinary care. Unlike the clean rebound in randomized withdrawal trials, the observed weight trajectories after discontinuation looked unexpectedly stable.²⁷ That is the story's necessary complication, and Figure 3 makes it visible.

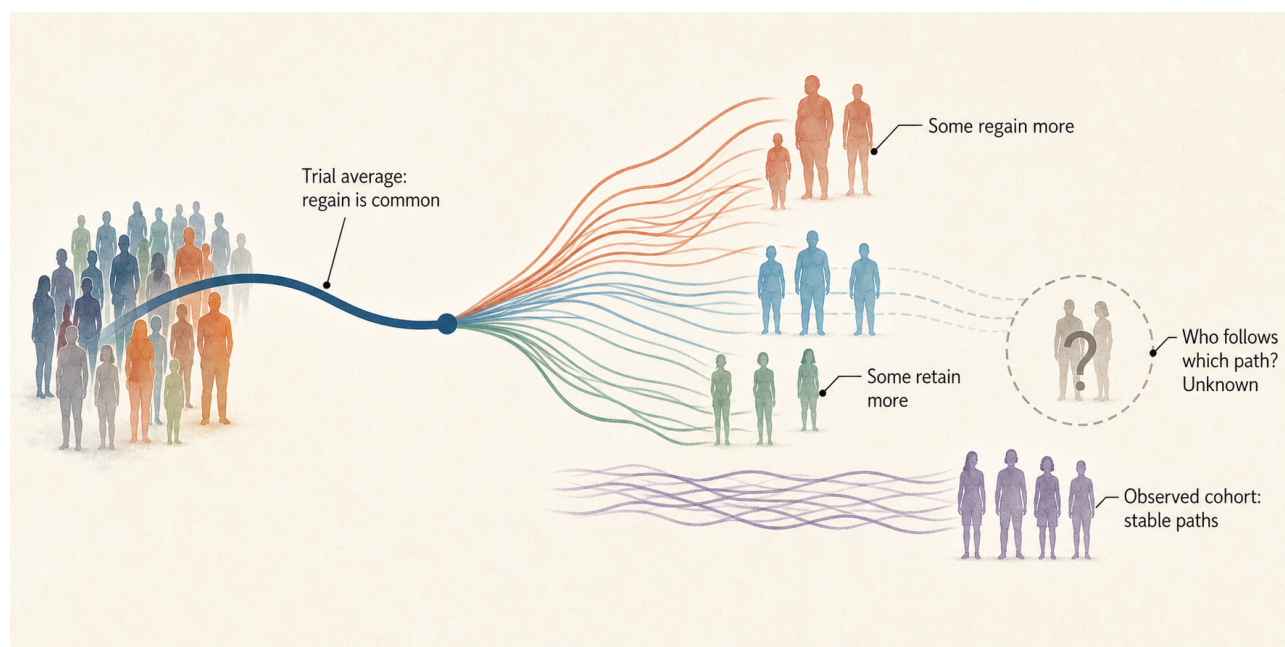


Figure 3. One mean curve can contain many personal paths. Randomized trials establish average regain after withdrawal, while selected participants retain substantial loss and one real-world cohort reported stable post-stop trajectories. Researchers still cannot reliably predict who follows which path.^{2,27,4,1,28}

The cohort does not overturn the trials. Pharmacy gaps are an imperfect marker of stopping, follow-up weights were sparse, doses were often lower, and people may have changed treatment or behavior in ways the records missed.²⁷ Yet the study exposes something the randomized averages cannot supply: a rule for the individual in front of the graph. Real-world reviews also find that stopping within the first year is common, so that missing rule affects many people.⁴ Withdrawal trials of semaglutide, liraglutide and tirzepatide agree about the average direction; they do not agree on your destination.^{1,28,29}

07 THE MAINTENANCE EXPERIMENT HAS BARELY BEGUN

Lundgren's team paired supervised exercise with the related drug liraglutide, then Jensen and colleagues followed participants after treatment ended. Those who had exercised held on to more of the benefit than those who had received liraglutide alone, and the combined program also beat the drug-only path after withdrawal.^{30,29,31} It is encouraging evidence for exercise as part of maintenance. It is not a semaglutide tapering trial.

No high-quality randomized study has yet shown that slowly stepping down semaglutide, staying on a lower dose, or adding ordinary behavioral support prevents rebound. Broad reviews have not found clear average protection from behavioral programs once medication stops, and Damhof's group treats tapering as a proposal still waiting for a proper test.^{15,13}

The evidence therefore points to a precise answer. When Ozempic's signal fades, your body usually does not enter a discrete withdrawal syndrome; it loses an active appetite- and glucose-regulating effect. Hunger can grow, weight often returns, and blood-pressure, lipid and glucose gains can narrow. Stomach symptoms may ease. The size and composition of the rebound, especially at Ozempic doses and in diabetes, remain personal questions the current trials cannot answer.

The final injection in Rubino's trial began the signal fading, but science still cannot predict its course through one body. A trial that compares abrupt stopping, gradual tapering and lower-dose maintenance—with the same support, repeated appetite and body-composition measurements, and years rather than months of follow-up—would finally show whether the dial can be held somewhere in between.

*
**

REFERENCES

31 · VERIFIED VIA CROSSREF

1. Rubino D, Abrahamsson N, Davies M, Hesse D, Greenway FL, Jensen C, Lingvay I, Mosenzon O, Rosenstock J, Rubio MA, Rudofsky G, Tadayon S, Wadden TA, Dicker D, STEP 4 Investigators, Friberg M, Sjödin A, Dicker D, Segal G, Mosenzon O, Sabbah M, Sofer Y, Vish-litzky V, Meesters EW, Serlie M, van Bon A, Cardoso H, Freitas P, Carneiro de Melo P, Monteiro M, Monteiro M, Rodrigues D, Badat A, Joshi P, Latiff G, Mitha EA, Snyman HH, van Nieuwenhuizen E, González Albarrán O, Caixas A, de al Cuesta C, García Luna PP, Morales Portillo C, Mezquita Raya P, Rubio MA, Abrahamsson N, Hoffstedt J, von Wöhrn F, Uddman E, Bach-Kliengel B, Beuschlein F, Bilz S, Golay A, Rudofsky G, Strey C, Fadieienco G, Kosei N, Tatarchuk T, Velychko V, Zynich O, Aronoff SL, Bays HE, Brockmyre AP, Call RS, Crump C, Desouza CV, Espinosa V, Free AL, Gandy WH, Geller SA, Gottschlich GM, Greenway FL, Han-Conrad L, Harper W, Herman L, Hewitt M, Hollander P, Kaster SR, Manassis A, Martin FA, McNeill RE, Murray AV, Norwood PC, Reed JCH, Rosenstock J, Rubino DM, Schear MJ, Warren ML (2021) Effect of Continued Weekly Subcutaneous Semaglutide vs Placebo on Weight Loss Maintenance in Adults With Overweight or Obesity. *JAMA*. <https://doi.org/10.1001/jama.2021.3224>
2. Wilding JPH, Batterham RL, Davies M, Van Gaal LF, Kandler K, Konakli K, Lingvay I, McGowan BM, Oral TK, Rosenstock J, Wadden TA, Wharton S, Yokote K, Kushner RF, STEP 1 Study Group (2022) Weight regain and cardiometabolic effects after withdrawal of semaglutide: The STEP 1 trial extension. *Diabetes, Obesity and Metabolism*. <https://doi.org/10.1111/dom.14725>
3. Tzang CC, Wu PH, Luo CA, Chen ZT, Lee YT, Huang ES, Kang YF, Lin WC, Tzang BS, Hsu TC (2025) Metabolic rebound after GLP-1 receptor agonist discontinuation: a systematic review and meta-analysis. *eClinicalMedicine*. <https://doi.org/10.1016/j.eclinm.2025.103680>
4. Thomsen RW, Mailhac A, Løhde JB, Pottegård A (2025) Real-world evidence on the utilization, clinical and comparative effectiveness, and adverse effects of newer GLP - 1RA -based weight-loss therapies. *Diabetes, Obesity and Metabolism*. <https://doi.org/10.1111/dom.16364>
5. Overgaard RV, Navarra A, Ingwersen SH, Bækdal TA, Kildemoes RJ (2021) Clinical Pharmacokinetics of Oral Semaglutide: Analyses of Data from Clinical Pharmacology Trials. *Clinical Pharmacokinetics*. <https://doi.org/10.1007/s40262-021-01025-x>
6. Blundell J, Finlayson G, Axelsen M, Flint A, Gibbons C, Kvist T, Hjørpsted JB (2017) Effects of once-weekly semaglutide on appetite, energy intake, control of eating, food preference and body weight in subjects with obesity. *Diabetes, Obesity and Metabolism*. <https://doi.org/10.1111/dom.12932>
7. Wharton S, Calanna S, Davies M, Dicker D, Goldman B, Lingvay I, Mosenzon O, Rubino DM, Thomsen M, Wadden TA, Pedersen SD (2022) Gastrointestinal tolerability of once-weekly semaglutide 2.4 mg in adults with overweight or obesity, and the relationship between gastrointestinal adverse events and weight loss. *Diabetes, Obesity and Metabolism*. <https://doi.org/10.1111/dom.14551>
8. Moiz A, Filion KB, Toutoumchi H, Tsoukas MA, Yu OHY, Peters TM, Eisenberg MJ (2025) Efficacy and Safety of Glucagon-Like Peptide-1 Receptor Agonists for Weight Loss Among Adults Without Diabetes. *Annals of Internal Medicine*. <https://doi.org/10.7326/annals-24-01590>
9. Saha B, Kamalumpundi V, Codipilly DC (2025) GLP1 and GIP Receptor Agonists: Effects on the Gastrointestinal Tract and Management Strategies for Primary Care Physicians. *Mayo Clinic Proceedings*. <https://doi.org/10.1016/j.mayocp.2025.09.017>
10. Friedrichsen M, Breitschaft A, Tadayon S, Wizert A, Skovgaard D (2021) The effect of semaglutide 2.4 mg once weekly on energy intake, appetite, control of eating, and gastric emptying in adults with obesity. *Diabetes, Obesity and Metabolism*. <https://doi.org/10.1111/dom.14280>
11. Gibbons C, Blundell J, Tetens Hoff S, Dahl K, Bauer R, Bækdal T (2021) Effects of oral semaglutide on energy intake, food preference, appetite, control of eating and body weight in subjects with type 2 diabetes. *Diabetes, Obesity and Metabolism*. <https://doi.org/10.1111/dom.14255>
12. Gabe MBN, Breitschaft A, Knop FK, Hansen MR, Kirkeby K, Rathor N, Adrian CL (2024) Effect of oral semaglutide on energy intake, appetite, control of eating and gastric emptying in adults living with obesity: A randomized controlled trial. *Diabetes, Obesity and Metabolism*. <https://doi.org/10.1111/dom.15802>
13. Damhof MA, van Bruggen FH, van Roon EN (2026) Early Weight Regain After GLP -1 Receptor Agonist Discontinuation: Mechanisms and Implications for Treatment De-Escalation Strategies. *Diabetes, Obesity and Metabolism*. <https://doi.org/10.1111/dom.71075>
14. Berg S, Stickle H, Rose SJ, Nemec EC (2025) Discontinuing glucagon-like peptide-1 receptor agonists and body habitus: A systematic review and meta-analysis. *Obesity Reviews*. <https://doi.org/10.1111/obr.13929>
15. West S, Scragg J, Aveyard P, Oke JL, Willis L, Haffner SJP, Knight H, Wang D, Morrow S, Heath L, Jebb SA, Koutoukidis DA (2026) Weight regain after cessation of medication for weight management: systematic review and meta-analysis. *BMJ*. <https://doi.org/10.1136/bmj-2025-085304>
16. Patel H, Babli SA, Shah S, Khambholja SH, Lingadahalli TD, Villarson C, Rizwanullah U, Kumarsivarajah S, Masood A, Rahming N, Drum Christie SS, Wallace LL, Alamu OS (2026) Post-cessation Weight Regain After Weight Management Medications: A Systematic Review and Meta-Analysis. *Cureus*. <https://doi.org/10.7759/cureus.111786>
17. Kolli RT, Aoutla S, Jyothi N, Mohamed Kalifa MRH, Raju A, Cheenikkal Muralidharan K (2025) Rebound or Retention: A Meta-Analysis of Weight Regain After the Discontinuation of Glucagon-Like Peptide-1 (GLP-1) Receptor Agonists and Other Anti-obesity Drugs. *Cureus*. <https://doi.org/10.7759/cureus.94926>

18. Cárdenas-Salas JJ, Sierra Poyatos RM, Luca BL, Sánchez Lechuga B, Modroño Móstoles N, Montoya Álvarez T, Gómez Montes MDLP, Ruiz Sánchez JG, Meneses González D, Sánchez-Lopez R, Casado Cases C, Pérez de Arenaza Pozo V, Vázquez Martínez C (2024) REAL life study of subcutaneous SEMaglutide in patients with type 2 diabetes in SPain: Ambispective, multicenter clinical study. Results in the GLP1-experienced cohort. *Journal of Diabetes and its Complications*. <https://doi.org/10.1016/j.jdiacomp.2024.108874>
19. McGowan BM, Bruun JM, Capehorn M, Pedersen SD, Pietiläinen KH, Muniraju HAK, Quiroga M, Varbo A, Lau DCW (2024) Efficacy and safety of once-weekly semaglutide 2.4 mg versus placebo in people with obesity and prediabetes (STEP 10): a randomised, double-blind, placebo-controlled, multicentre phase 3 trial. *The Lancet Diabetes & Endocrinology*. <https://doi.org/10.1016/s2213-8587%2824%2900182-7>
20. McCrimmon RJ, Catarig AM, Frias JP, Lausvig NL, le Roux CW, Thielke D, Lingvay I (2020) Effects of once-weekly semaglutide vs once-daily canagliflozin on body composition in type 2 diabetes: a substudy of the SUSTAIN 8 randomised controlled clinical trial. *Diabetologia*. <https://doi.org/10.1007/s00125-019-05065-8>
21. Karakasis P, Patoulas D, Fragakis N, Mantzoros CS (2025) Effect of glucagon-like peptide-1 receptor agonists and co-agonists on body composition: Systematic review and network meta-analysis. *Metabolism*. <https://doi.org/10.1016/j.metabol.2024.156113>
22. Batsis JA, Gavras A, Gross DC, Cheever CR, Da Silva BR, Meira Filho LF, Jones EP, Batchek D, Khandpekar S, Patel R, Awkal B, Pape A, Bahna M, Zamboni M, Prado CM (2026) Effect of Incretin-Based and Nonpharmacologic Weight Loss on Body Composition. *Annals of Internal Medicine*. <https://doi.org/10.7326/annals-25-00478>
23. Eisa N, Barood O (2026) Lean Mass Changes With Incretin Therapy Versus Lifestyle Intervention: A Systematic Review and Meta-Analysis of Randomised Controlled Trials. *Diabetes, Obesity and Metabolism*. <https://doi.org/10.1111/dom.70666>
24. Ryan DH (2025) New drugs for the treatment of obesity: do we need approaches to preserve muscle mass? *Reviews in Endocrine and Metabolic Disorders*. <https://doi.org/10.1007/s11154-025-09967-4>
25. Locatelli JC, Costa JG, Haynes A, Naylor LH, Fegan PG, Yeap BB, Green DJ (2024) Incretin-Based Weight Loss Pharmacotherapy: Can Resistance Exercise Optimize Changes in Body Composition? *Diabetes Care*. <https://doi.org/10.2337/dci23-0100>
26. Bhandarkar A, Bhat S, Kapoor N (2025) Effect of GLP-1 receptor agonists on body composition. *Current Opinion in Endocrinology, Diabetes & Obesity*. <https://doi.org/10.1097/med.0000000000000934>
27. Gasoyan H, Butsch WS, Schulte R, Casacchia NJ, Le P, Boyer CB, Griebeler ML, Burguera B, Rothberg MB (2025) Changes in weight and glycemic control following obesity treatment with semaglutide or tirzepatide by discontinuation status. *Obesity*. <https://doi.org/10.1002/oby.24331>
28. Aronne LJ, Sattar N, Horn DB, Bays HE, Wharton S, Lin WY, Ahmad NN, Zhang S, Liao R, Bunck MC, Jouravskaya I, Murphy MA, SUR-MOUNT-4 Investigators, Fretes JO, Coronel MJ, Gutnisky LL, Frechtel GD, Gellersztejn E, Aizenberg D, Maldonado N, Pereira M, Santos QG, Calil Salim C, Canani LH, Halpern B, Russo LA, Bodart J, Neto DV, Augusto G, Leite S, Yang YC, Lin WY, Huang CN, Huang KC, Fitz-Patrick D, Pau CH, Toth PD, Freeman GH, Gardner DF, Wynne AG, Loy J, Horn DB, Mehra PK, Layle S, Berghold JH, de Souza J, Nadar VK, Albizu Angulo GR, Cohen KR, Smith TR, Aronne LJ, Vaughn M, Alcantara-Gonzalez AA, Forman SB, Agaiby JM, Geller SA, Fraser NJ, Jenders RA, Barbel-Johnson KM, Mayfield RK, Vance CD, Prier KT, Murray AV, Lillestol MJ, Denham DS, Park JY, Klein EJ, Bays HE, Philis-Tsimikas A, Bressler PE, Reed JC, Aslam S, Rosenstock J, Frias JP, Klaff LJ, Brazg R, Gomez-Cuellar M, Connery L, Van JT, Selam JL, Kim J, Blake D, Gabriel J, Arora S, McCartney MJ, Solano RK, Brodie SK, Nardandrea JP (2024) Continued Treatment With Tirzepatide for Maintenance of Weight Reduction in Adults With Obesity. *JAMA*. <https://doi.org/10.1001/jama.2023.24945>
29. Jensen SBK, Blond MB, Sandsdal RM, Olsen LM, Juhl CR, Lundgren JR, Janus C, Stallknecht BM, Holst JJ, Madsbad S, Torekov SS (2024) Healthy weight loss maintenance with exercise, GLP-1 receptor agonist, or both combined followed by one year without treatment: a post-treatment analysis of a randomised placebo-controlled trial. *eClinicalMedicine*. <https://doi.org/10.1016/j.eclinm.2024.102475>
30. Lundgren JR, Janus C, Jensen SBK, Juhl CR, Olsen LM, Christensen RM, Svane MS, Bandholm T, Bojsen-Møller KN, Blond MB, Jensen JEB, Stallknecht BM, Holst JJ, Madsbad S, Torekov SS (2021) Healthy Weight Loss Maintenance with Exercise, Liraglutide, or Both Combined. *New England Journal of Medicine*. <https://doi.org/10.1056/nejmoa2028198>
31. Jensen SBK, Janus C, Lundgren JR, Juhl CR, Sandsdal RM, Olsen LM, Andresen A, Borg SA, Jacobsen IC, Finlayson G, Stallknecht BM, Holst JJ, Madsbad S, Torekov SS (2022) Exploratory analysis of eating- and physical activity-related outcomes from a randomized controlled trial for weight loss maintenance with exercise and liraglutide single or combination treatment. *Nature Communications*. <https://doi.org/10.1038/s41467-022-32307-y> ■